

Circular decoding + a final shifted prediction

Ordinary sampling builds the scene. One additional prediction sees the wrap as an interior region. Circular decoding reconstructs the final pixels.

Ordinary sampling / native decode



Ordinary sampling / circular 128



Final shifted prediction / circular 128



Full ERP context: same temple frame 97. The wrap is at the left/right edges. The next page centres it in the view.

Where it touches

At the **final denoising step**, compute the ordinary prediction. Roll the in-progress video latent by half its width, predict again at the same noise level, then roll that prediction back. Accept only its first and last latent columns. Keep the ordinary prediction elsewhere and for audio.

Complete the original final Euler update. Wrap opposite-edge context around the finished video latent, run the native VAE decoder, then crop the added margins. The shift covers the **whole clip**, not its last video frame.

Tested recipe

One latent column per edge (16 output pixels) | 128 px decoding context per edge | LoRA 1.0 | H3 BF16 | VAE FP16

Incremental gain beyond circular decoding alone

Scene	Output / frames	Steps	Control MAE	Final-shift MAE	Reduction
Temple	1536 x 672 / 243	50	4.0981	3.3319	18.70%
Forest	1536 x 672 / 243	50	8.3747	7.6123	9.10%
Temple	1024 x 448 / 124	100	5.3703	3.7191	30.75%

Boundary MAE: absolute RGB difference between the first and last columns, averaged over rows, channels and every frame (0-255). These percentages are pixel-error reductions, **not perceived-quality improvements**. Each row compares the same circular-128 decoder.

Cost of the method: one extra model evaluation (50 -> 51 or 100 -> 101). No new training or weight changes. The extra prediction took about 27 s in the native temple H200 test; loading and decoding are separate.

Native-resolution temple: inspect the join

1536 x 672 | 243 frames | 50 steps | frame 97. Identical projections of verified PNGs. No sharpening or generated enhancement.

Context / 75-degree view / eye level

Native decoder



Circular decoder 128



Circular 128 + final shifted prediction



Close-up / 20-degree view / eye level

Native decoder



Circular decoder 128



Circular 128 + final shifted prediction



Context / 75-degree view / looking up 45 degrees

Native decoder



Circular decoder 128



Circular 128 + final shifted prediction



100 steps: the final placement works best

1024 x 448 | 124 frames | same temple seed. Each variant starts from the same saved late trajectory; the circular-128 decoder stays fixed.

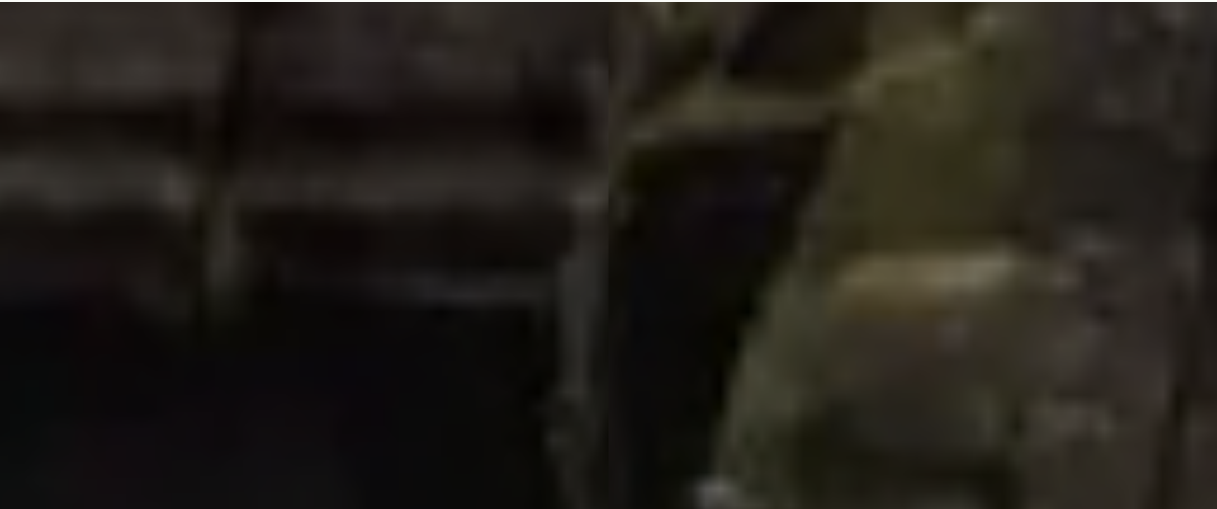
Correction step	Sigma	Ordinary steps afterward	Boundary MAE	Reduction
No correction	-	-	5.3703	-
95	0.433735	5	5.1559	3.99%
99	0.196721	1	4.6393	13.61%
100 (final)	0.108108	0	3.7191	30.75%

Step 99 uses the same noise level as the 50-step schedule's final step. Final placement still wins here. Sigma, update size and subsequent predictions change together; their individual contributions are not isolated.

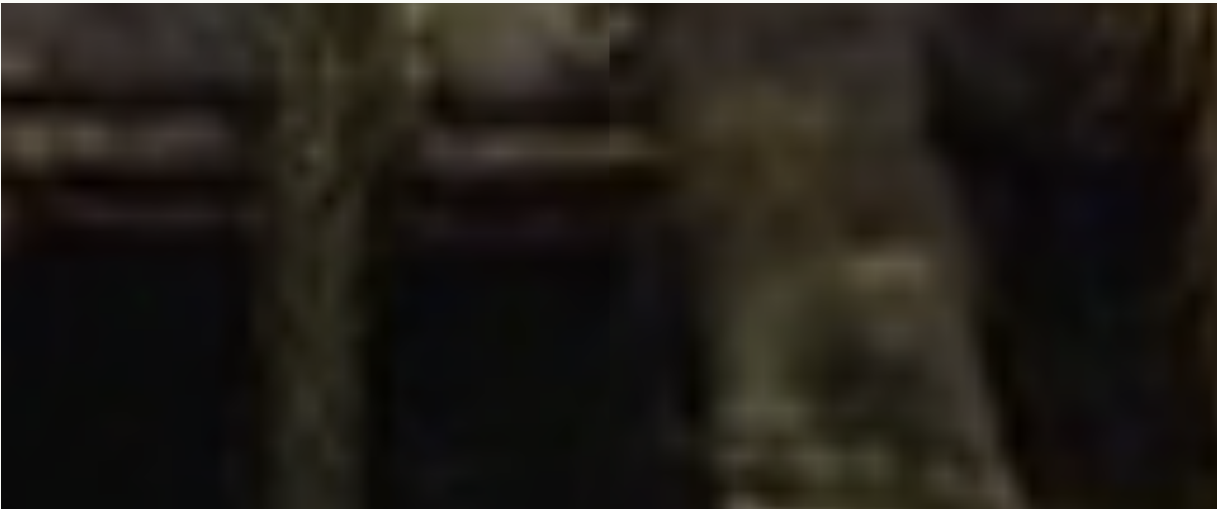
Circular 128 / ordinary sampling / frame 0



Circular 128 / correction at step 100 / frame 0



Circular 128 / ordinary sampling / frame 123



Circular 128 / correction at step 100 / frame 123



30-degree views centred on the wrap. Residual structural mismatch remains; pixel matching alone does not establish an invisible seam.

Forest transfer and the current boundary of evidence

1536 x 672 | 243 frames | 50 steps | frame 97. A second scene: the additional boundary-error reduction is 9.10%, with some foliage softness.

Circular 128 / ordinary sampling



Seam-centred 75-degree view



Verified: exact canonical final-step counterfactuals; unchanged exterior video latent and audio; repeated 100-step control matches every decoded frame. Layout is retained in inspected views. Residual seams and local texture changes remain.

Circular 128 / final shifted prediction



Seam-centred 75-degree view



Still open: more independent scenes, 100 steps at native resolution, image conditioning, refinement/upscaling, long continuation and poles. This report does not establish those workflows.

Method, settings, image hashes and limitations: github.com/ShamanicArts/h3-spherical-vae